



**K S Tharunpranav 251801359-2025-AIDS** ▾

**K2**

|              |                                   |
|--------------|-----------------------------------|
| Started on   | Thursday, 27 August 2026, 8:22 AM |
| State        | Finished                          |
| Completed on | Monday, 31 August 2026, 11:49 AM  |
| Time taken   | 4 days 3 hours                    |
| Marks        | 1.00/1.00                         |
| Grade        | 10.00 out of 10.00 (100%)         |

**Question 1** | Correct | Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten  $i$  burgers with  $c$  calories each, then he has to run at least  $3^i * c$  kilometers to burn out the calories. For example, if he ate 3

burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are  $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$ .

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance

he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm. Apply greedy approach to solve the problem.

**Input Format**

First Line contains the number of burgers

Second line contains calories of each burger which is  $n$  space-separate integers

**Output Format**

Print: Minimum number of kilometers needed to run to burn out the calories

**Sample Input**

```
3
5 10 7
```

**Sample Output**

```
76
```

**For example:**

| Test        | Input      | Result |
|-------------|------------|--------|
| Test Case 1 | 3<br>1 3 2 | 18     |

**Answer:** (penalty regime: 0 %)

```

1 | #include <stdio.h>
2 | #include <stdlib.h>
3 | int compare(const void *a, const void *b) {
4 |     return (*(int *)b - *(int *)a);
5 | }
6 | int main() {
7 |     int n;
8 |     scanf("%d", &n);
9 |     int calories[n];
10 |    for (int i = 0; i < n; i++) {
11 |        scanf("%d", &calories[i]);
12 |    }
13 |    qsort(calories, n, sizeof(int), compare);
14 |    long long total = 0;
15 |    long long power = 1;
16 |    for (int i = 0; i < n; i++) {
17 |        total += calories[i] * power;
18 |        power *= n;
19 |    }
20 |    printf("%lld\n", total);
21 |    return 0;
22 | }
```

|   | Test        | Input        | Expected | Got |   |
|---|-------------|--------------|----------|-----|---|
| ✓ | Test Case 1 | 3<br>1 3 2   | 18       | 18  | ✓ |
| ✓ | Test Case 2 | 4<br>7 4 9 6 | 389      | 389 | ✓ |
| ✓ | Test Case 3 | 3<br>5 10 7  | 76       | 76  | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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